

Character Identity and Discrete Crystal-Lattice Addressing in Topological Geometric Intelligence

Emylton Leunufna

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No affiliation

Abstract

Topological Geometric Intelligence (TGI) treats a raw character stream as an ordered sequence of persistent identities whose occurrences can acquire distinct geometric addresses. This manuscript defines the identity substrate, the distinction between a character ID and its event occurrence, a discrete five-coordinate frame layout, and the conditions under which traversal may reconstruct an output. The TGI instantiation maps each admitted Unicode scalar to $ord(c)+16$, reserves 0–15 for structural uses, rejects surrogate values, and does not normalize input. This differs from the normalized registry instantiation described in the companion NINMENI NMU preprint while preserving its one-character/one-identity axiom. A two-stream acceptance experiment places the repeated prefix *SISTEM* into separate context lanes. The specified queries reconstruct each complete raw sentence and return no path for an unformed third anchor. The outcome follows from a complete route certificate, not from an assumed semantic value of a coordinate or a probabilistic guess. Checkpoint-order falsifiers show why the original event sequence is part of the addressing contract. The manuscript separates the established exact-route result from the later tasks of forming knowledge and referents.

1. Research question and contribution

The NINMENI NMU paper [1] fixes identity granularity. TGI begins where that paper stops: if two experiences contain the same character and even the same initial phrase, how can a geometric memory keep their occurrences distinct and read a justified continuation? The answer must preserve the axiom *one Unicode character -> one static ID*; it may not create a new identity for *SISTEM*, replace positions by a pooled vector, or change identity according to frequency. The problem is one of **addressing and route validity**, not a new tokenization scheme.

This manuscript contributes (a) a typed separation of identity, occurrence, frame, coordinate, and route; (b) a concrete discrete placement law that separates source lanes; (c) a complete-route versus absent-route decision; and (d) falsifiers that distinguish correct text output from correct geometric state. These responsibilities are upstream of band and phase formation in the companion phase manuscript. A route can be well-formed even when its text is false about the world; world-relation evidence is owned by the interventional relations manuscript.

2. Identity and occurrence definitions

Let U be the declared admitted Unicode scalar space. The TGI codec is $I: U \rightarrow N$, $I(c)=ord(c)+16$. The low 16 integers are not character IDs. For admitted scalars, the map is injective and static. A surrogate is rejected before it can become a stored identity. Input is neither NFC-normalized nor segmented. This is an explicit TGI instantiation choice. In [1], NINMENI's measured corpus and 10,240-slot registry use a declared normalization and hash-bound table; importing that registry's implementation into TGI would be an unjustified architectural shortcut. What TGI takes from [1] is the identity **axiom**, not its corpus-specific table.

For a source s , frame f , and position i , define the occurrence $o=(s,f,i,I(c_i))$. The projection $identity(o)=I(c_i)$ forgets event ownership, so two occurrences can share an identity without being interchangeable. The ordered stream is $O_f=(o_0, \dots, o_{\{T-I\}})$; its chronology and raw source must survive checkpointing. A valid bulk-native operator may process several occurrences together, but it must project back to every original identity and position in the same order. This is a stronger invariant than equality of the final rendered string.

The distinction matters for repeated letters and shared prefixes. In the sentence *SISTEM TGI ...*, both *S* occurrences have the same static ID but not the same occurrence address. The *S* at the start of the other sentence also has that ID, while its source lane differs. Neither a numerical ID nor a single coordinate carries a meaning label such as *safe* or *reactor*.

3. Discrete layout and route certificate

The versioned foundation layout places occurrence i of the n th distinct frame at

$$P(n,i)=(2+i, 1, 0, 3+5n, (5-3n) \bmod 11).$$

The first coordinate orders characters; the fourth and fifth create source lanes. This is a constructive address allocation, not a proof that Euclidean or hyperbolic geometry by itself forms knowledge. For an admitted frame, a forward adjacency bond joins $P(n,i)$ to $P(n,i+1)$ only within its own lane. The route certificate records source, ordered occurrence identities, addresses, bond validity, and an explicit terminal marker. A reader returns the original character stream only after reaching the terminal marker through checked bonds. Three outcome categories are kept distinct:

Outcome	Necessary observation	Interpretation
<i>COMPLETE</i>	every ordered occurrence and terminal bond verified	exact reconstruction of a stored route
<i>NO_PATH</i>	requested anchor is absent from valid state	no certified continuation
<i>INCOMPLETE</i>	a route is present but broken or stopped by a ceiling	no complete output claim

The codec can decode each visited ID back to its scalar. A complete certificate therefore implies exact output character by character for that stored route. It does not imply factual truth of the decoded sentence. The no-path rule is a decision about the current geometric memory, not a universal statement that a sentence is impossible.

4. Method and falsifiers

The frozen user acceptance case contains two raw streams:

1. *SISTEM TGI AMAN DARI HALUSINASI*;
2. *SISTEM REAKTOR NUKLIR MEMBUTUHKAN PENDINGIN*.

The two streams share the first six characters. The accepted anchors are the first *S* in lane A and the first *S* in lane B; a third unformed anchor is an absence control. Successful evaluation requires exact reconstruction of both strings, zero cross-lane jumps after *SISTEM*, and no output from the absent anchor. The acceptance report [5] executes the route and the broader project regression in the same chain.

Output equality is not sufficient to verify spatial ownership. A historical checkpoint fault sorted sources before replay, producing the same text while changing which frame obtained which lane. The spatial retention decision [6] therefore freezes the original raw insertion journal and verifies the lattice after fresh-process recovery. Frame equality is checked from the complete NMU sequence rather than trusting a potentially colliding digest. A broken bond, modified coordinate, or mismatched receipt must fail validation rather than fall back to a plausible text completion.

The acceptance test is deliberately sharp: shared prefixes make a shallow character lookup appear sufficient for the first few positions, but only a lane-bound route can preserve the correct continuation. Conversely, an absent anchor should not be mapped to the closest observed source. The reader has no softmax sampling step that could hide absence behind an output string.

5. Results and inference

The source acceptance report [5] records both exact reconstructions and the absent-context decision. The checkpoint-order falsifier and fix are retained in [6]. The phase-validity requirements required before a frame contributes are defined separately in [7]; they prevent an invalid source receipt from silently serving as a route. The source distribution report [8] checks that the relevant implementation and tests can be reproduced from extracted package bytes. These evidence classes should not be collapsed: a regression checks behavior, a checkpoint check tests state transport, and a source archive check tests reproducibility.

This establishes the TGI substrate and exact route behavior for the declared states. It does not claim that $P(n,i)$ alone discovers a physical referent or that the original integral expression for xi in *KONSEP.txt* has been derived from this layout. That expression is a source hypothesis. The tested placement law is an explicit implementation whose premises and consequences are visible.

6. Reproducibility and artifact statement

The authoritative source is *KONSEP.txt* [4], the four NINMENI reference preprints, the acceptance contract and raw receipt [5], source-preserving checkpoint decision [6], and the hash-bound source distribution [8]. The relevant implementation entry points are *tgi/identity.py*, *tgi/spatial.py*, *tgi/frame_engine.py*, and their tests in the distribution. The publication bundle identifies exact SHA-256 bytes for the manuscript, source, and evidence. No training checkpoint or private corpus is required to reproduce the acceptance case.

7. Formal invariants and a constructive argument

The character mapping is injective because adding a fixed offset to two different Unicode scalar numbers cannot make them equal. This is an identity property, not a statistical observation. The occurrence projection is not injective: two events may share $I(c)$. Thus no proof about event ownership may be based on ID equality alone. The address allocation must additionally bind the frame number, within-frame position, and source journal. For two frames $n \neq m$, their fourth coordinates differ by $5(n-m)$; hence their lanes cannot coincide under the declared layout. For two positions in the same frame, first coordinates differ. This establishes occurrence separation inside the admitted finite frame set independently of the wording of the sentences. The fifth coordinate is an additional consistency check, not the only source of uniqueness.

For exact reconstruction, take the route whose bonds connect the ordered occurrences $o_0, \dots, o_{(T-1)}, END$. The base case reads $I(c_0)$ from o_0 . At step $i < T-1$, the validated forward bond has the same frame lane and the unique successor $P(n, i+1)$; decoding it appends $c_{(i+1)}$. Induction gives the original scalar sequence after T occurrences, provided the terminal bond is present. If a bond is missing, the induction premise fails and the route cannot emit a complete result. This is the scope of the constructive theorem. It does not establish that a different unstored sentence should have the same result.

The absence decision is equally precise. An unformed anchor has no initial valid occurrence and therefore no initial route certificate. Returning an empty result is a property of the current state; returning one of the two stored sentences would be a counterexample to the contract. A safety ceiling interrupting a nonempty route is recorded as *INCOMPLETE*, avoiding the mistake of conflating unobserved data with a proof of absence.

8. Test matrix and state provenance

Probe	Perturbation	Required certificate consequence
same prefix, two sources	trigger first S under each lane	exact respective full stream, no cross-lane edge
third anchor	request a lane never formed	<i>NO_PATH</i> , no guessed phrase
source order swap on recovery	reorder journal while keeping text	reject changed coordinate ownership

Probe	Perturbation	Required certificate consequence
damaged bond or terminal	remove one required route edge	<i>INCOMPLETE</i> or invalid certificate, never partial text labeled complete
Unicode distinction	feed different scalar sequences with similar display	preserve distinct IDs and raw order

The journal and the certificate carry different roles. The journal is provenance of what entered and when; the certificate is a derivation from that journal under current rules. A certificate recovered without the underlying raw history would be insufficient to test whether source order changed. Conversely, merely retaining the journal does not prove that the current lattice still obeys the placement law. Both are checked after recovery. This mirrors the NINMENI papers' distinction between a declared identity rule and a measured state, while keeping TGI's own discrete addressing rule explicit.

Code and artifact availability

The public TGI runtime and selected regression tests are available at <https://github.com/rfls-emyton/TGI/tree/de95c4c> and as the *tgi-foundation* version *0.2.0.dev0* distribution at <https://pypi.org/project/tgi-foundation/0.2.0.dev0/>. The GitHub commit fixes the code revision used for this publication series. Paths under *evidence/*, the historical *KONSEP.txt* and *HIPOTESIS* archive, and other unpublished workspace reports identify research provenance retained by the author; they are not files in the public GitHub repository or PyPI distribution. Results requiring those receipts are therefore identified by their archived source paths rather than presented as independently downloadable from the code release.

References

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